Dual coding

Using diagrams and words together

Use words and visuals together when a diagram makes a relationship, sequence or structure easier to understand. Select the representation for its explanatory purpose.

How to do it well

1 Choose the relationship. Decide what the visual will make clearer: position, sequence, cause, comparison or structure.

2 Design the representation. Use simple shapes, consistent conventions and labels placed close to the relevant features.

3 Explain in stages. Point or trace while introducing each important part, keeping spoken language aligned with the visual.

4 Allow processing. Pause for reading, annotation or a short explanation; do not introduce another demand while students inspect new information.

5 Rehearse the meaning. Ask students to interpret or reconstruct the visual and explain the relationship in words before using it in a new task.

What this sounds like

“Follow this arrow as I explain the transfer. Now describe what it represents. Which part of the diagram shows where the energy is stored?”

Classroom example. In Science, a simple circuit diagram appears in stages as the teacher traces the route. Labels are placed beside components and reading time is separated from narration. Pupils then explain what a connection represents and complete a missing part. The diagram carries the relationship; decorative images would add no useful information.

Top tips that make the technique work

Use a meaningful visual. A diagram should explain the content rather than decorate the slide.

Put labels near their referent. Students should not have to search between a distant key and the relevant feature.

Build gradually. Reveal a structure in the same order as the explanation so attention follows the intended sequence.

Avoid competing language. Pause narration while students read substantial new text.

Ask students to use the representation. Labelling, tracing, explaining and recreating reveal more than simply copying the picture.

Do and avoid

Do	Avoid
Align speech, pointing and the displayed feature.	Using decorative pictures as evidence of dual coding.
Remove visuals that add no explanatory value.	Crowding the diagram with every possible label at once.

Further reading

Practitioner guidance; teacher language above is an illustrative BSCS adaptation.

DistillED - Dual Coding